United International University

Department of Computer Science & Engineering

CSE 3715: Data Communications

Class Test 03

Duration: 30 Minutes | Total Marks: 20

1. Given the data polynomial:

$$D(x) = x^7 + x^6 + x^4 + x^2 + 1$$

and the generator polynomial:

$$G(x) = x^3 + x + 1$$

   (a) Perform CRC encoding and write the final transmitted message (data appended with CRC bits). **[7]**

   (b) Assuming no error occurred during transmission, show how the message from 1(a) is decoded in the receiver end. **[7]**

2. A message has been received in the form of six 6-bit blocks (including the checksum block):

101100 110011 000111 111000 011010 101000

Decode the message using checksum to check whether the message has been received **error-free** or not. **[6]**